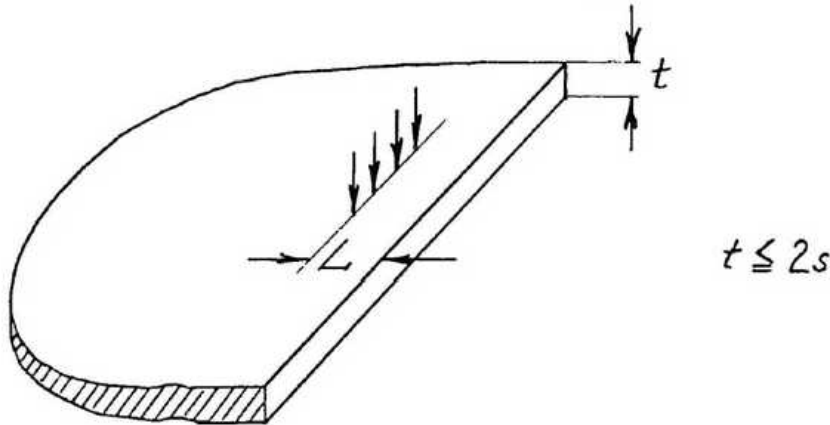


E.5) Probe Array Parallel to Edge, Thin Sample.

When the sample is thin, it is convenient to write the resistivity in the following way:

$$\rho = G \frac{V}{I}, \quad G = \frac{\pi}{\ln 2} \cdot t \cdot D_4\left(\frac{L}{s}\right) \cdot F_4\left(\frac{t}{s}, \frac{L}{s}\right) \quad (16)$$

Where

$\frac{\pi}{\ln 2} \cdot t = 4.5324 \cdot t$ is the geometric factor for an infinite slice of thickness $t \ll s$,

$$D_4\left(\frac{L}{s}\right) = \frac{1}{1 + \frac{1}{2\ln 2} \cdot \ln \left[\frac{(L/s)^2 + 1}{(L/s)^2 + \frac{1}{4}} \right]} \quad \text{is the additional} \quad (17)$$

correction to apply when measuring at a distance L from the straight edge on the semi-infinite slice of thickness $t \ll s$.

$F_4\left(\frac{t}{s}, \frac{L}{s}\right)$ deviates from unity, when the thickness t of the slice is not much less than the probe distances s .

The expression for $D_4\left(\frac{L}{s}\right)$ was obtained from the formula (23) section I.3. for a circular slice when the probes are perpendicular to a diameter, by letting the diameter go to infinity. $D_4\left(\frac{L}{s}\right)$ is tabulated and plotted at page 29.

The additional correction $F_4\left(\frac{t}{s}, \frac{L}{s}\right)$ to apply when t is not much

less than s , was computed on the basis of Uhlir's paper (f) (g) and tabulated below. Curves for $F_4\left(\frac{t}{s}, \frac{L}{s}\right)$ are shown at page 30.